The impact of post-trial tasks on brain activation during real-time sentence-level semantic composition

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Introduction. Experimental studies of sentence processing usually employ post-trial tasks, but whether the tasks should directly target the linguistic aspect of interest or be orthogonal to it remains unresolved, though task demands have been suggested to involve top-down semantic control [1,2]. This fMRI study aims to unravel how the nature of post-trial tasks affects neural activation during combinatorial semantic computation. We examine underspecified meaning composition (e.g., (1) "*The frog hopped for 5 minutes/5 months.*") versus its semantically explicit counterpart (e.g., (2) "*… swam for 5 minutes.*"). Whereas the former induces an iterative reading that is morphosyntactically unsupported, the latter yields a transparent one-time reading. The process of semantic specification for (1) demands a lexically-driven conceptual search through context, which elicits extra cost privileging the left anterior cortex [3,4,5]. We hypothesize that (i) computing underspecified meaning increases cost and brain recruitment. And given interaction to the conceptual system, (ii) different tasks will selectively impact semantic computation by directing attention to different aspects of meaning construal, thereby modulating neural responses through top-down *semantic control*. Our comparison set is [3] that used an iterativity judgment task (**ITER**), in which participants were asked *how many times* the denoted event occurred; it reports greater activation in the left inferior frontal gyrus (LIFG) and parietal regions. Adopting the same stimuli, we used naturalness-ratings (**NAT**) as the post-trial task, and conducted joint analyses comparing the ITER with NAT responses. While both tasks require full sentence comprehension, ITER directly taps the target meaning of event iterativity whereas NAT is orthogonal to it.

Method. The two fMRI experiments used the same 32 sets of 4 conditions: *NoIter*, *PuncIter*, *DurIter*, and *GapIter* in Japanese (Table 1), along with a distinct set of 80 fillers. All stimuli were normed for naturalness and interpretations via a pre-test questionnaire (n=22). Native Japanese speakers were recruited to read sentences and perform either naturalness ratings (*1: very unnatural ~ 4: very natural*) in the NAT experiment (n=30-5), or iterativity judgments for the denoted event (*once, 2~10 times, 11~100 times, > 101 times*) in the ITER experiment (n=20-2). The procedures and paradigms were identical across experiments (TR = 1 sec, TE = 30 ms). Data analysis targeted the sentence-reading window, excluding task execution period. Whole-brain analysis identified 14 ROIs commonly activated for the *semantically-underspecified (PuncIter, DurIter, GapIter) > transparent (NoIter)* contrast across experiments. We further compared regional activation for ITER vs. NAT per ROI to assess task differences.

Findings. (i) Computing underspecified meaning (> transparent baseline) elicited greater activation in the LIFG, its right homologue, and left intraparietal sulcus (Fig. 1), reflecting a deeper conceptual-contextual search [3,4,5]. **(ii)** In the commonly activated ROIs, neural responses were larger for ITER than NAT, with task differences particularly pronounced in the bilateral frontal regions (Fig. 2). Processing underspecified meaning requires a costly meaning search supported by the fronto-parietal network. Beyond core semantic computation, a task that directly probes the target meaning (ITER) engages a more robust search with greater neural activation, compared to a task that assesses sentences' "gestalt" semantic representation (NAT). In line with cognitive control approaches to meaning composition [6], results suggest that both semantic content and task-dependent cognitive allocation shape sentence processing profiles. Tasks taping into the

core comprehension processes enhance brain activation compared to tasks remaining orthogonal to them, a fact that can be exploited to optimize experimental sensitivity.

Table 1: Sample stimuli of the 4 critical conditions shared by the two experiments (NAT & ITER)

Condition	Sample sentence
(A) NolIter Durative verb + Short interval	*kaeru-ga 20 pun-kan ike-de oyoi-da-to kodomo-wa haha-ni shabet-ta.* frog-NOM 20 minute for pond-LOC swim-PAST-COMP child-TOP mother-DAT talk-PAST 'The child told the mother that the frog **swam for 20 minutes** in the pond.'
(B) PunclIter Punctual verb + Short interval	*kaeru-ga 20 pun-kan ike-de haneta-da-to kodomo-wa haha-ni shabet-ta.* frog-NOM 20-minute for pond-LOC jump-PAST-COMP child-TOP mother-DAT talk-PAST 'The child told the mother that the frog **jumped for 20 minutes** in the pond.'
(C) DurIter Durative verb + Long interval	*kaeru-ga 2 kagetsu-kan ike-de oyoi-da-to kodomo-wa haha-ni shabet-ta.* frog-NOM 2-month for pond-LOC swim-PAST-COMP child-TOP mother-DAT talk-PAST 'The child told the mother that the frog **swam for 2 months** in the pond.'
(D) GapIter Punctual verb + Long interval	*kaeru-ga 2 kagetsu-kan ike-de haneta-da-to kodomo-wa haha-ni shabet-ta.* frog-NOM 2-month for pond-LOC jump-PAST-COMP child-TOP mother-DAT talk-PAST 'The child told the mother that the frog **jumped for 2 months** in the pond.'

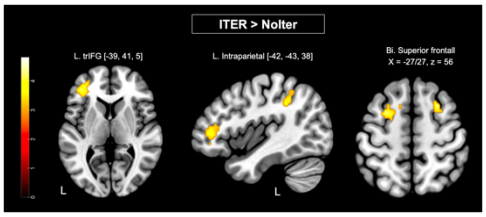

Fig. 1. Results of whole-brain analysis with joint datasets. ITER = {PunclIter, DurIter, GapIter} with underspecified iterative meaning combined.

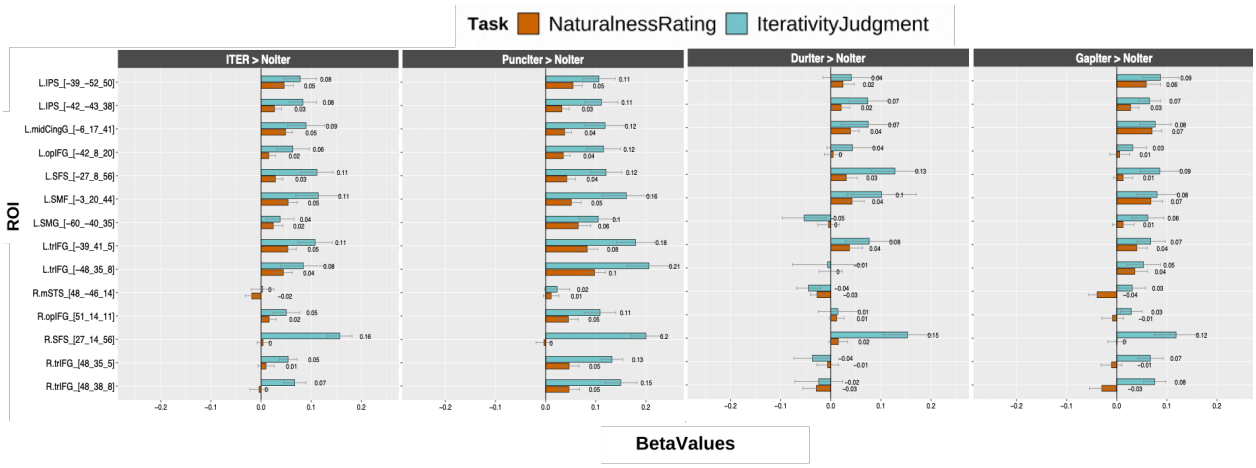

Fig. 2. ROI results of task comparisons showing regional activation level per contrast.

Abbreviations: L = left hemisphere; R = right hemisphere; IFG = inferior frontal gyrus; tr = pars triangularis; op = pars opercularis; mid = middle; SFS = superior frontal sulcus; SMF = superior middle frontal regions; SMG = supramarginal gyrus; Inf = inferior. STS = superior temporal sulcus

References. [1] Jackson, R. L. (2021). The neural correlates of semantic control revisited. *NeuroImage, 224*, Article 117444. **[2]** Graves et al. (2010). Neural correlates of implicit and explicit combinatorial semantic processing. *Neuroimage*, *53*(2), 638-646. **[3]** Lai et al. (2023). Neural underpinnings of processing combinatorial unstated meaning and the influence of individual cognitive style. *Cereb. Cortex*, *33*(18), 10013-10027. **[4]** Deo & Piñango (2011, Sep.). Quantification and context in measure adverbs. In *SALT* (pp. 295-312). **[5]** Piñango et al. (2024). Comprehension of English *for*-adverbials: The Nature of Lexical Meanings and the Neurocognitive Architecture of Language. *TiCS.* **[6]** Lambon Ralph et al. (2017). The neural and computational bases of semantic cognition. *Nat. Rev. Neurosci*, *18*(1), 42-55.